Patuakhali Science and Technology University
Faculty of Computer Science and Engineering

CIT 320 :: Software Development Project-II

Project Proposal

Project Title : QuizAid
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Contents

1. Abstract 3
2. Objectives 3
3. Problem Statement 3
4. Related Work 3
5. Scope 3
    5.1. Job Market Analysis 4
6. Methodology 5
    6.1. Technology Stack 5
    6.2. Design Principles 5
7. Visual Models 5
    7.1. ERD (Entity Relationship Diagram) 6
    7.2. System Architecture Diagram 7
    7.3. Timeline (Gantt Chart) 7
8. UI Mockups 8
9. Future Plans 9
10. Result 9
11. References 10
12. Weekly Report 11

QuizAId
ACADEMIC QUIZ

1. Abstract

QuizAid is a lightweight quiz platform that lets teachers create online quizzes and share a Zoom-style **join link / ID + passcode** so students can attempt from mobile. This version of QuizAid will be built using **Flutter (Dart)** for the client app and **Firebase** services (Authentication + Firestore/Realtime DB) for backend and real-time features. It targets Bangladeshi education contexts where connectivity can be variable and teachers need instant feedback without complex setup.

2. Objectives

- Enable quiz creation and sharing via join link/ID + passcode.
- Provide frictionless student attempts (optional name/roll capture), timer, autosave, auto-grade.
- Deliver actionable analytics (per-student, per-item, per-class) with export.
- Ensure security, privacy, and reliability.

3. Problem Statement

Teachers lack simple, share-by-link assessments that run on low-end devices and low bandwidth many existing tools feel heavy or require payments. Students rarely receive item-level feedback quickly, so data for remediation is missing. Schools also need a privacy-aware solution that works on widely available Android devices and is easy to deploy and maintain.

4. Related Work

- **Kahoot!** – Fun quizzes but limited academic performance tracking.
- **Quizizz** – Strong quiz games, limited classroom interaction.
- **Google Classroom** – Strong class management, lacks native quiz hosting.
- **Toppr** – Quiz content but mainly paid and region-focused.
- **Socrative** – Feature-rich but not fully optimized for mobile-first classroom experience.

5. Scope

• **Dynamic Quiz Engine:** Support for multiple question formats, including Multiple Choice (MCQs), True/False, and short-answer questions.

• **Real-Time Assessment:** Instant scoring and feedback mechanisms that provide students with correct answers and explanations immediately after submission.

• **Timed Examination:** Integrated countdown timers for individual questions or entire quiz sessions to simulate real-world exam conditions.

• **Performance Dashboard:** A personalized user profile that tracks quiz history, accuracy rates, and progress over time using visual charts.

• **Secure Authentication:** User registration and login system to ensure progress and data are securely stored and synced across devices.

5.1. Job Market Analysis

Job Portal	Position / Link
BD Jobs	• Flutter Developer (Mobile App): https://bdjobs.com/jobs/?keyword=flutter • Mobile Application Developer (Flutter & Firebase): https://bdjobs.com/jobs/?keyword=flutter%20firebase
BD Tech Jobs & TechTalents	• Software Engineer – Flutter: https://bdtechjobs.com/jobs?search=flutter • Junior/Mid Flutter Developer: https://bdtechjobs.com/jobs?search=mobile%20developer
eJobs Bangladesh & JobMatchingBD	• Android/iOS Developer (Flutter): https://www.ejobs.com.bd/jobs/?q=flutter • Mobile App Developer (Cross-Platform): https://jobmatchingbd.com/job-category/it-jobs/
Arc.dev & Wellfound	• Flutter Mobile Developer (Remote): https://arc.dev/remote-jobs/flutter • Cross-Platform App Developer: https://wellfound.com/jobs?keyword=flutter
Careerjet Bangladesh	• Flutter & Firebase App Developer: https://www.careerjet.com.bd/flutter-jobs.html • Mobile Software Engineer: https://www.careerjet.com.bd/mobile-developer-jobs.html

6. Methodology

6.1. Technology Stack

Frontend (Client-Side)

- **Framework: Flutter** – For building a high-performance, natively compiled application from a single codebase.
- **Language: Dart** – Optimized for UI development with strong null safety.
- **Local Storage: Hive** or **Isar** – Lightweight, lightning-fast NoSQL databases for caching quiz data and user progress offline.

Backend

- **Authentication: Firebase Auth** – Supports Email/Password, Google, and Social logins with built-in security.
- **Primary Database: Cloud Firestore** – A NoSQL document database for storing quiz content, user profiles, and real-time leaderboard rankings.
- **Real-time Logic: Firebase Cloud Functions** – For server-side logic (e.g., calculating final scores securely to prevent cheating).
- **File Storage: Firebase Storage** – To host media-heavy quiz content like images, audio clips, or explanatory videos.

6.2. Design Principles

The design of QuizAid will adhere to the following principles:

- **Platform-Agnostic UI (Adaptive Design):** Leveraging Flutter's layout engine to deliver a pixel-perfect experience that adapts fluidly across **iOS, Android, and Web** from a single codebase.
- **Optimistic UI Updates:** Using **Firebase Real-time Sync** to reflect user actions (like answering a question) immediately on the interface before the server confirms, ensuring a "zero-lag" feel.
- **Data-Lean Content Loading:** Implementing lazy-loading and image caching to ensure that even users on low-bandwidth connections experience fast question transitions.

7. Visual Models

7.1. ERD (Entity Relationship Diagram)

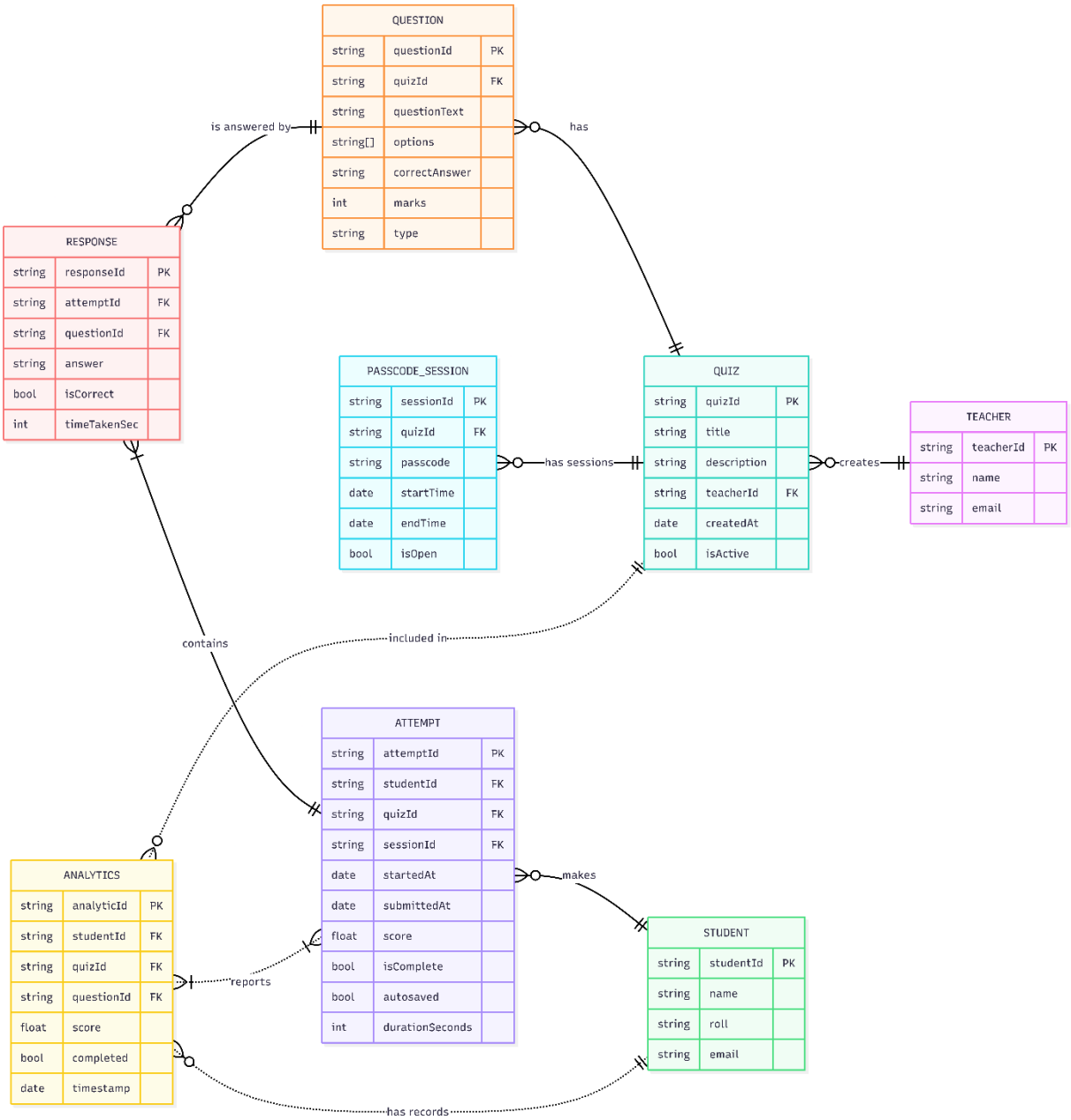


Figure 1: Entity Relationship Diagram of QuizAid

7.2. System Architecture Diagram

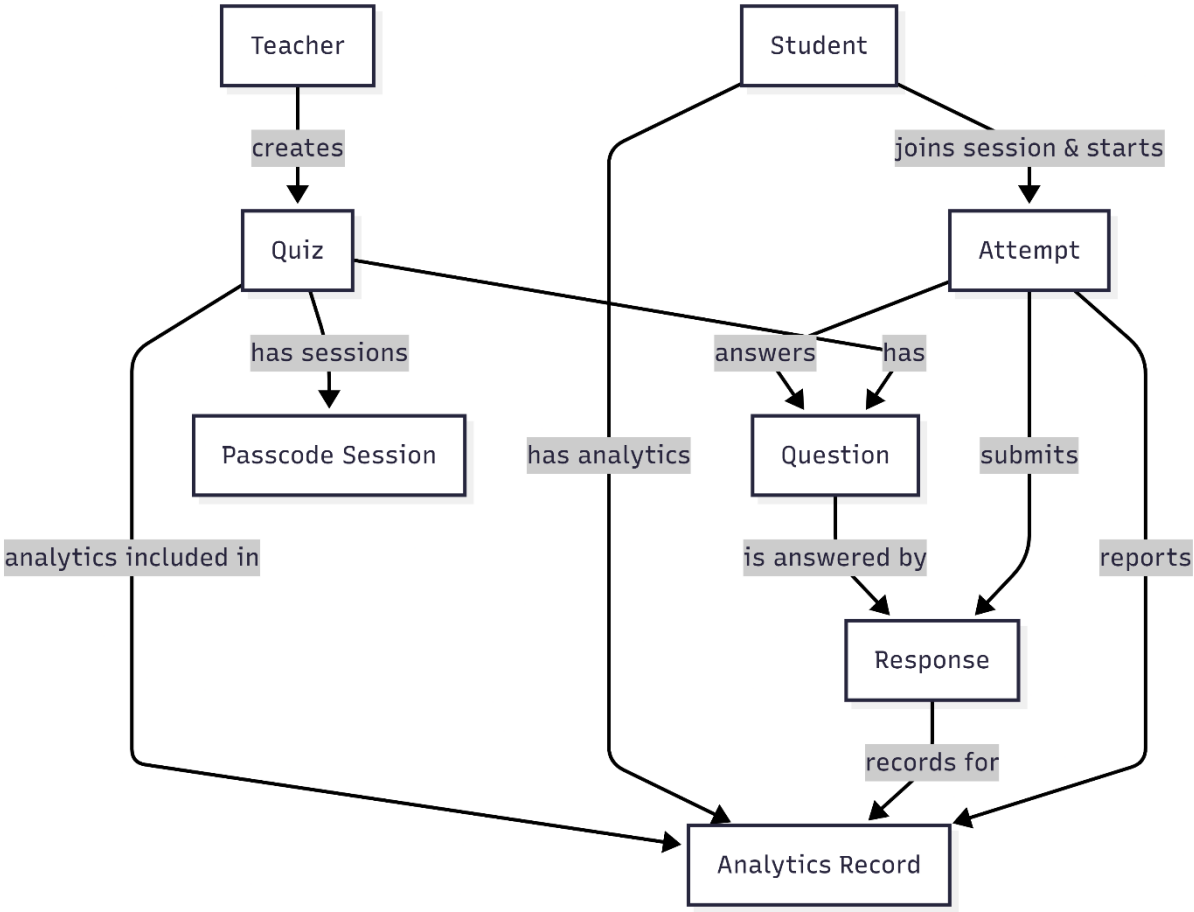


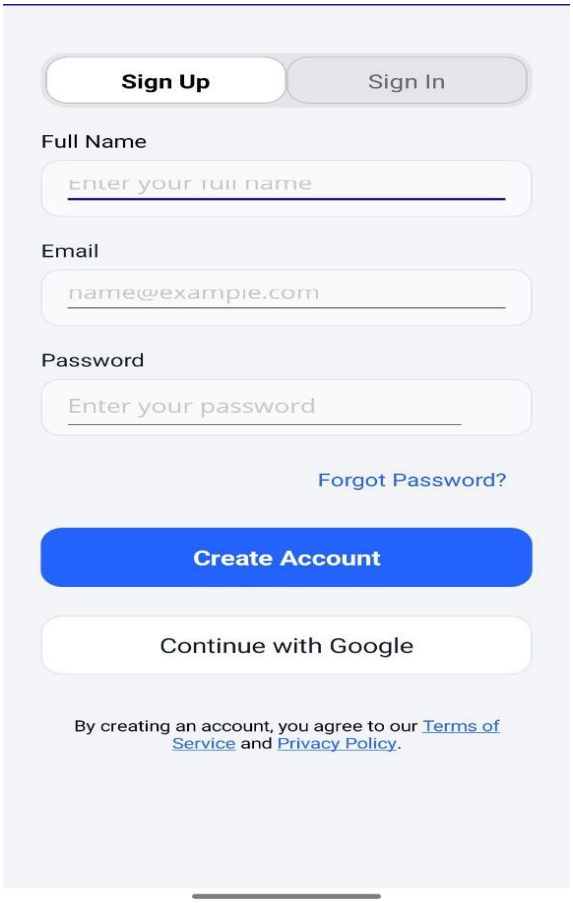
Figure 2: Flow charts

7.3. Timeline (Gantt Chart)

The timeline is divided into 12 weeks, focusing on the development lifecycle.

Task	**Week 1-2**	**Week 3-4**	**Week 5-6**	**Week 7-8**	**Week 9**	**Week 10-11**	**Week 12**
UI Mockups + Firebase Project Setup	✓						
Authentication & Classroom Module		✓					
Quiz Creation & Practice Mode			✓				
Live Quiz + Leaderboard				✓			
Analytics + Testing					✓		
UI Polish + Documentation						✓	
Final Deployment (APK/Web)							✓

8. UI Mockups



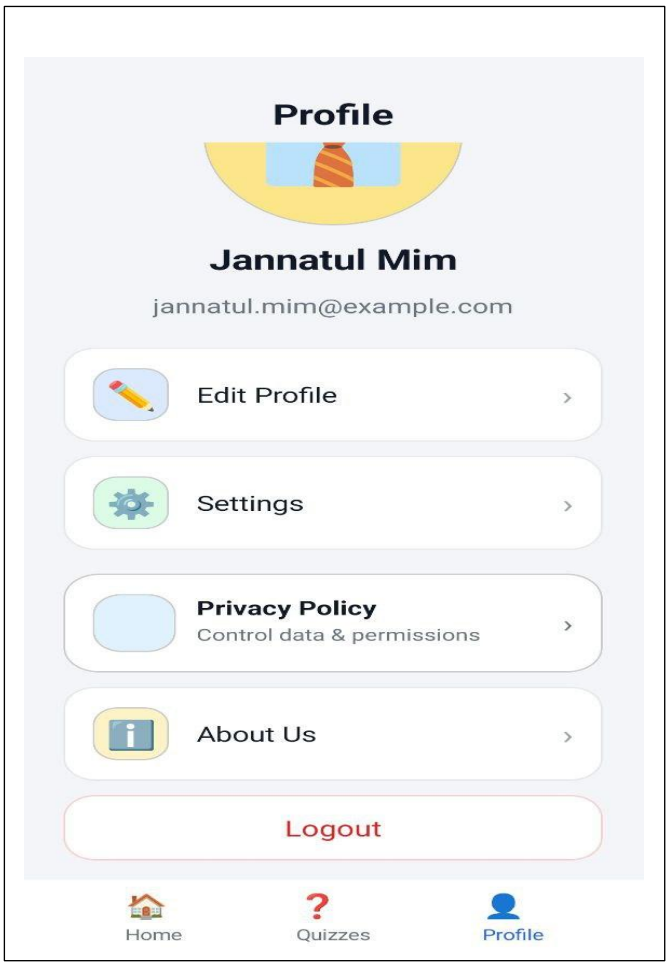


Fig: Sign up & Profile page

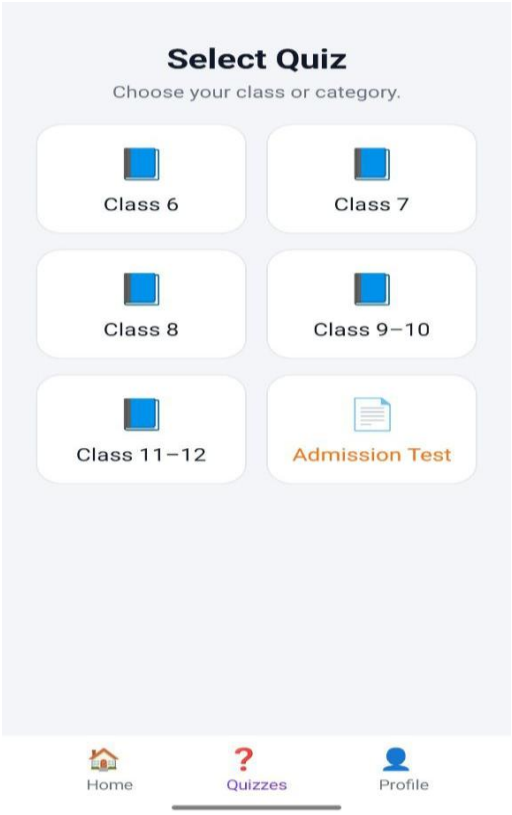


Fig: Quizzes and Home Page

9. Future Plans

Time-boxed scope: only core features (authoring, join-by-link, attempts, basic analytics) are guaranteed; advanced analytics and full offline sync are deferred. Bandwidth constraints: optimized for low data, but extremely poor connectivity may still affect join/autosave/submission.

Academic integrity controls: basic shuffle/timer/passcode only; no proctoring or advanced anti-cheat.

10. Result

QuizAid aims to deliver a responsive, real-time mobile app that improves student-teacher interaction and provides structured learning through quizzes and analytics.

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[5] “Firebase Authentication for Flutter.” Accessed: Jan. 26, 2026. [Online]. Available: https://firebase.google.com/docs/auth/flutter/start

[6] “Riverpod State Management.” Accessed: Jan. 24, 2026. [Online]. Available: https://riverpod.dev/docs/introduction/getting_started

[7] “Flutter Material 3 Design.” Accessed: Jan. 22, 2026. [Online]. Available: https://m3.material.io/develop/flutter

12. Weekly Report

No.	Date	Objective	Status	Remarks
1		Define core functionality of QuizAid, including login/signup for student and teacher models.		
2		Set up all and connect Firebase		
3		Authentication Flow: Implements email/password login and registration with role-based navigation for teachers and students.		
4		Classroom Management: Allows teachers to create classrooms, add students, and manage class lists.		
5		Exam Management: Enables teachers to create exams with custom questions, schedule them, and assign to students.		
6		Student Exam & Progress Tracking: Students can take assigned exams; app tracks scores and progress, displaying updates to teachers.		
7		Practice Quizzes: Implements practice quizzes for students based on academic classes and subjects, with instant feedback and answer review.		
8		Recheck and Review: Students can review their answers and explanations for completed quizzes and exams to enhance learning.		
9		Create profile and add all the features in profile to ensure the facilities for users		
10		Testing and Deployment: Conducts app testing, security checks, offline support, and prepares for Android/iOS deployment.		